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Shortest Distance

Define - Shortest Distance [Book 142 page]

When two lines do not intersect and are not parallel i.e. they do not lie in the same plane, then these lines are ~~perpendicular~~ said to be non-intersecting lines. The straight line which is perpendicular to each of these two non-intersecting lines is called the line of shortest distance [and the length of this line intercepted between the given lines is called the length of the shortest distance.]

Find the shortest distance between two given lines and also to obtain the equations of the shortest distance.

Answer:- From Book - [142-143 page]

Figure 3.12

2(a) 7

Find the length of the shortest distance between the lines

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \; ; \; \frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$$

Find also its equations and the points where it intersects the lines.

Solution:- Given lines

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \quad - - - (1)$$

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \quad - - - (2)$$

Let the shortest distance meet the given lines in L and M.

Any point on (1) suppose L is $(r+3, -2r+5, r+7)$

Any point on (2) suppose M is $(7r'-1, -6r'-1, r'-1)$

Direction ratio's of LM are $[(x_2-x_1, y_2-y_1, z_2-z_1)]$

$$(7r'-r-4, -6r'+2r-6, r'-r-8) \quad - - - (3)$$

Since LM is perpendicular to both the given lines, then

$$1(7r'-r-4) - 2(-6r'+2r-6) + 1(r'-r-8) = 0$$

$$7(7r'-r-4) - 6(-6r'+2r-6) + 1(r'-r-8) = 0$$

Solving this we get, $r=0, r'=0$

Therefore, the points where the shortest distance intersects the lines (1) and (2) i.e. L is $(3,5,7)$ and M is $(-1,-1,-1)$.

Now the length of the shortest distance is

$$LM = \sqrt{(-1-3)^2 + (-1-5)^2 + (-1-7)^2}$$

$$= \sqrt{16+36+64}$$

$$= \sqrt{116} = 2\sqrt{29}.$$

Again, direction ratios of LM are

$4, 6, 8$ or $2, 3, 4$. [From (3), putting $r=r'=0$]

Hence the equation of the shortest distance i.e. LM is

$$\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$$

L(3,5,7)
DR's of LM
(2,3,4)
∴ Equation of SD
$\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$

—X—

\# Find the length and the equation of the shortest distance between the lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \; ; \; \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}.$$

Solution:- Given lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad - - - (1)$$

$$\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5} \quad - - - (2)$$

Let (l, m, n) be the direction cosines of the shortest distance of the lines so that it is perpendicular to both the given lines. Then

$$2l + 3m + 4n = 0$$
$$3l + 4m + 5n = 0$$

$$\therefore \frac{l}{15-16} = \frac{m}{12-10} = \frac{n}{8-9}$$

$$\Rightarrow \frac{l}{-1} = \frac{m}{2} = \frac{n}{-1} = \frac{1}{\sqrt{6}} \quad \left[\frac{1}{\sqrt{(-1)^2 + 2^2 + (-1)^2}} = \frac{1}{\sqrt{6}} \right]$$

Hence direction cosines of shortest distance are $\left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$

Let a point on the 1st line is $B(1,2,3)$

and a point on the 2nd line is $D(2,4,5)$.

Therefore, the length of the shortest distance

$$= l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

$$= \frac{-1}{\sqrt{6}}(2-1) + \frac{2}{\sqrt{6}}(4-2) - \frac{1}{\sqrt{6}}(5-3)$$

$$= -\frac{1}{\sqrt{6}} + \frac{4}{\sqrt{6}} - \frac{2}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{6}}.$$

Now, the equation of the plane containing the line (1) and the line of shortest distance is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ -1 & +2 & -1 \end{vmatrix} = 0$$

$$\Rightarrow 11x + 2y - 7z + 6 = 0$$

Again, the equation of the plane containing the line (2) and the line of shortest distance is

$$\begin{vmatrix} x-2 & y-4 & z-5 \\ 3 & 4 & 5 \\ -1 & 2 & -1 \end{vmatrix} = 0$$

$$\Rightarrow 7x + y - 5z + 7 = 0.$$

Hence, the equation of the shortest distance is

$$11x+2y-7z+6=0,\ 7x+y-5z+7=0$$

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For Practise

1. What do you mean by shortest distance between two straight lines? Find the shortest distance between the lines (2018)

$$\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4} \text{ and } \frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$$

2. Find the equation of the shortest distance line between the lines (2017) $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$.

3. Find the shortest distance between the lines (2016) $\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$ and $\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$

4. Find the shortest distance between the lines (2015) $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$.

ACTIVE

Date :

Page :

Variable Plane

Remember:- Distance between a point and a plane:-

If a plane equation is given by $ax+by+cz+d=0$ and (m,n,p) be any point, then the distance between them is given by

$$d = \frac{am+bn+cp+d}{\sqrt{a^2+b^2+c^2}}$$

**Variable plane equation :- $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$.

Find the distance of the points $(2,0,1)$ and $(3,-3,2)$ from the plane $x-2y+z=6$.

Solution:- The equation of the plane is

$$x-2y+z=6$$

$$\Rightarrow x-2y+z-6=0$$

Its distance from $(2,0,1)$ is

$$d = \frac{2-2.0+1-6}{\sqrt{(1)^2+(-2)^2+(1)^2}}$$

$$= \frac{-3}{\sqrt{6}}$$

Its distance from $(3,-3,2)$ is

$$d = \frac{3-2.(-3)+2-6}{\sqrt{1^2+(-2)^2+1^2}}$$

$$= \frac{5}{\sqrt{6}}$$

——x——

(showed)

——x——

A variable plane is at a constant distance p from the origin and meets the axes in A, B and C. Show that the locus of the centroid of the tetrahedron OABC is

$$x^{-2}+y^{-2}+z^{-2}=16p^{-2}$$

Solution:- Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

$$\Rightarrow \frac{x}{a}+\frac{y}{b}+\frac{z}{c}-1=0 \quad - - - (1)$$

If its distance from the origin is P, then

$$p=\frac{0+0+0-1}{\sqrt{1/a^2+\frac{1}{b^2}+\frac{1}{c^2}}}$$

$$\Rightarrow \frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2} \quad - - - (2)$$

If (x,y,z) be the centroid of the tetrahedron OAB where the points are

$O(0,0,0)$, $A(a,0,0)$, $B(0,b,0)$, $C(0,0,c)$

then

$$x=\frac{0+a+0+0}{4}=\frac{a}{4}$$

$$\Rightarrow a=4x$$

Similarly, we get

$$y=\frac{b}{4} \quad \text{or, } b=4y$$

$$z=\frac{c}{4} \quad \text{or, } c=4z$$

Now putting the values of a, b, c in (2), we get the locus of the centroid which is

$$\frac{1}{p^2} = \frac{1}{16x^2} + \frac{1}{16y^2} + \frac{1}{16z^2}$$

$$\Rightarrow x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$$

(showed)

~~Ans (c)~~

A variable plane is at a constant distance 3p from the origin and meets the axes in A, B and C. Show that the locus of the centroid of the triangle ABC is

$$x^{-2} + y^{-2} + z^{-2} = p^{-2}.$$

Solution:- Let the equation of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} - 1 = 0 \quad - - - (1)$$

If its distance from the origin is 3p, then

$$3p = \frac{0+0+0-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\Rightarrow \frac{1}{9p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad - - - (2)$$

If (x,y,z) be the centroid of the triangle ΔABC where the points are $A(a,0,0)$, $B(0,b,0)$, $C(0,0,c)$

then $x = \frac{a+0+0}{3} \Rightarrow x = \frac{a}{3}$

or $a = 3x$

Similarly, we get

$$y = \frac{b}{3} \quad \text{or} \quad b = 3y$$

$$z = \frac{c}{3} \quad \text{or} \quad c = 3z$$

Now putting the values of a, b, c in (2), we get the locus of the centroid of the triangle which is

$$\frac{1}{9p^2} = \frac{1}{9x^2} + \frac{1}{9y^2} + \frac{1}{9z^2}$$

$$\Rightarrow x^{-2} + y^{-2} + z^{-2} = p^{-2}$$

(showed)

Now from a point outside a plane only one perpendicular can be drawn and hence d_1 and d_2 must be the same. Therefore, we get

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$$

(Proved)

———X———

A variable plane passes through a fixed point (a,b,c) and meets the axes of reference A, B and C. Show that the locus of the point of intersection of the planes through A, B, C parallel to the co-ordinate planes is

$$ax^{-1} + by^{-1} + cz^{-1} = 1.$$

Solution:- Let the plane be

$$\frac{x}{m} + \frac{y}{n} + \frac{z}{p} = 1 \quad \text{---} \quad (1)$$

If it passes through (a,b,c), then

$$\frac{a}{m} + \frac{b}{n} + \frac{c}{p} = 1 \quad \text{---} \quad (2)$$

Therefore it meets the axes in

$A(m,0,0)$, $B(0,n,0)$ and $C(0,0,p)$

Also planes through A, B and C are parallel respectively to the co-ordinate planes are

$$x = m, \quad y = n, \quad z = p \quad \text{---} \quad (3)$$

respectively.

Thus the locus of the point of intersection of the planes is obtained by eliminating the variables m, n, p.

From (2) and (3), we get

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$$

$$\Rightarrow ax^{-1} + by^{-1} + cz^{-1} = 1$$

(showed)

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A variable plane at a constant distance p from the origin meets the axes in A, B and C. Through A, B, C planes are drawn parallel to the co-ordinate planes, show that locus of their point of intersection is

$$x^{-2} + y^{-2} + z^{-2} = p^{-2}$$

Solution:- Let the equation of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad - - - (1)$$

Therefore it meets the axes in A(a,0,0), B(0,b,0) and C(0,0,c). Now its distance from origin is p given by

$$p = \frac{-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\text{or, } \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad - - - (2)$$

Also planes through A, B and C are parallel respectively to the co-ordinate planes are

$$x = a, \; y = b, \; z = c \quad - - - (3)$$

respectively.

Thus the locus of point of intersection of the planes is obtained by eliminating a, b, c. ~~From~~ (2). From (2) and (3), we get

$$\frac{1}{p^2} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}$$

$$\Rightarrow x^{-2} + y^{-2} + z^{-2} = p^{-2}.$$

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